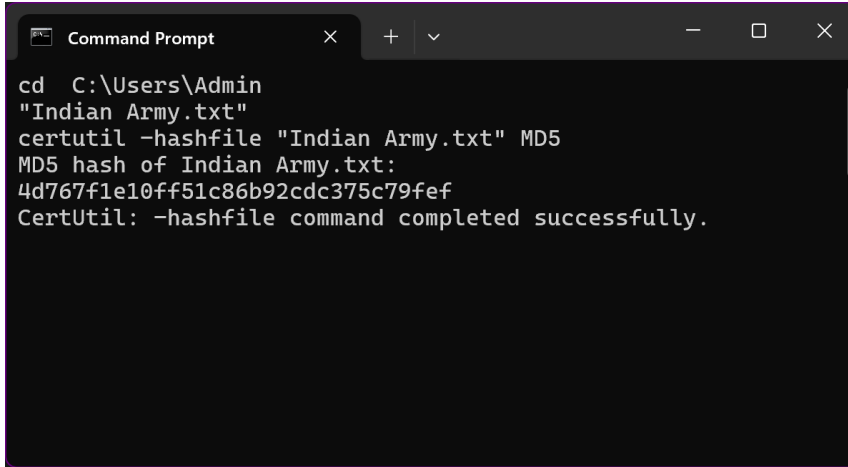


Lab 1 DFIR : File Hashing

-Eshan Shende : K027



```
Command Prompt
cd C:\Users\Admin
"Indian Army.txt"
certutil -hashfile "Indian Army.txt" MD5
MD5 hash of Indian Army.txt:
4d767f1e10ff51c86b92cdc375c79fef
CertUtil: -hashfile command completed successfully.
```

Generated Hash : 4d767f1e10ff51c86b92cdc375c79fef

A **file hash** is a unique, fixed-length alphanumeric string generated by a cryptographic hash function (such as MD5, SHA-1, or SHA-256) based on the content of a file. The hash value represents the specific data in the file, so any change in the file's content, even a single bit, will result in a completely different hash.

File hashes are important for various reasons, including:

1. **Data Integrity Verification:**
 - Ensures that a file has not been altered during transmission or storage.
 - Any tampering with the file would produce a different hash value.
2. **Authentication:**
 - Verifies the authenticity of a file, ensuring it comes from a trusted source and has not been compromised.
3. **Efficiency:**
 - Hashes allow large amounts of data to be summarized in a small string, making it easier to compare files without transferring or analyzing the entire file.

In digital forensics, file hashing plays a crucial role in ensuring the integrity and authenticity of digital evidence. Here's how:

1. Evidence Integrity Verification

- File hashes are used to prove that digital evidence has not been tampered with from the moment it was collected.

2. Deduplication

- During the analysis of digital evidence, many files might be identical copies. Hashing helps quickly identify duplicate files to streamline the investigation process.

3. Authentication of Evidence

- Hash values serve as a digital "fingerprint" for evidence. Courts require assurance that the evidence presented is the same as what was collected, and hashes provide this assurance.

